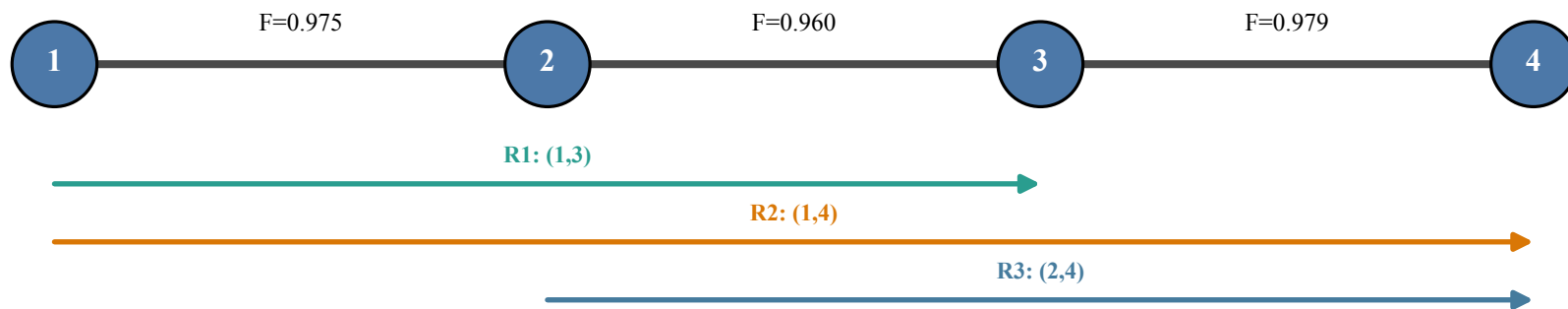


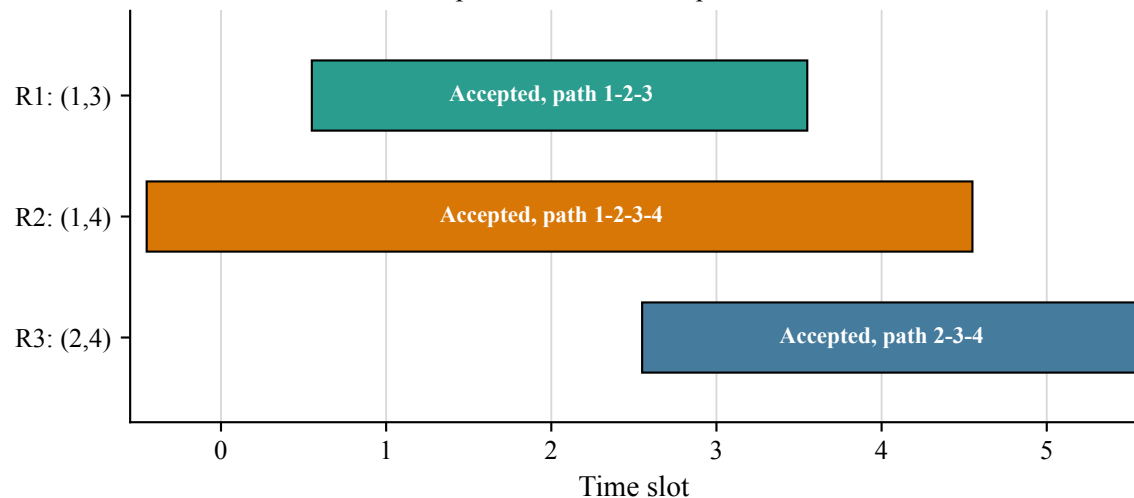
Small-scale optimality validation of WPFA

1. Known small instance: a 4-node chain with three requests

$T=6$, memory=4 per node, fidelity threshold=0.80



2. Exact optimum: all three requests are scheduled



R2 uses one purification round on link (3,4).

3. WPFA stays very close to the true optimum

Exact OPT	2.072766
WPFA	2.071154
Optimality gap	0.078%

*Both methods accept all 3 requests.
Exact search: 1,584 schedules enumerated, 900 feasible, 197 B&B states.
No approximation, candidate cap, or time cutoff was used for OPT.*